



Artificial Intelligence and Machine Learning

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AI Landscape

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Types of Agents

- Simple Reflex Agents
- Model-Based Reflex Agents
- Goal-Based Agents
- Utility-Based Agents
- Learning Agents

Simple Reflex Agent

- Action is based only on the current percept
- No history or internal state
- Works well in fully observable environments
- Example: Thermostat: Turns heating on/off based on current temperature

Model-based Reflex Agent

- Maintains an internal state of the environment
- Uses models to handle partially observable environments
- Example: Vacuum robot remembers areas it has already cleaned

Goal-based Agent

- Select actions that achieve specific goals
- Requires planning and decision-making
- Example: Route-finding in Google Maps to reach a destination

Utility-based Agent

- Chooses actions based on utility value (how desirable an outcome is)
- Can handle conflicting goals and uncertainties
- Example: Investment AI choosing between high-return and low-risk options

Learning Agent

- Capable of improving performance over time through learning
- Has four components:
 1. Learning element
 2. Critic
 3. Performance element
 4. Problem generator
- Example: Game AI improving with each round played (e.g., AlphaGo)

Activity

Identify types of agents used in:

- Siri
- Amazon Alexa
- Netflix Recommendations
- Chess-playing bots



THANK YOU

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